HARVARD MEDICAL SCHOOL

Dana-Farber Cancer Institute

# Relatives’ similarity in test score performance is explained by a wide range of genetic and environmental parameters

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**Based on:** Eftedal NH, Eilertsen EM, Sunde HF, Kleppestø TH, Ystrom E, Czajkowski NO (2025). Beyond additive genetic effects: Explaining family resemblance in school performance across millions of pairs of Norwegian relatives. *PNAS* 122(25):e2419627122.

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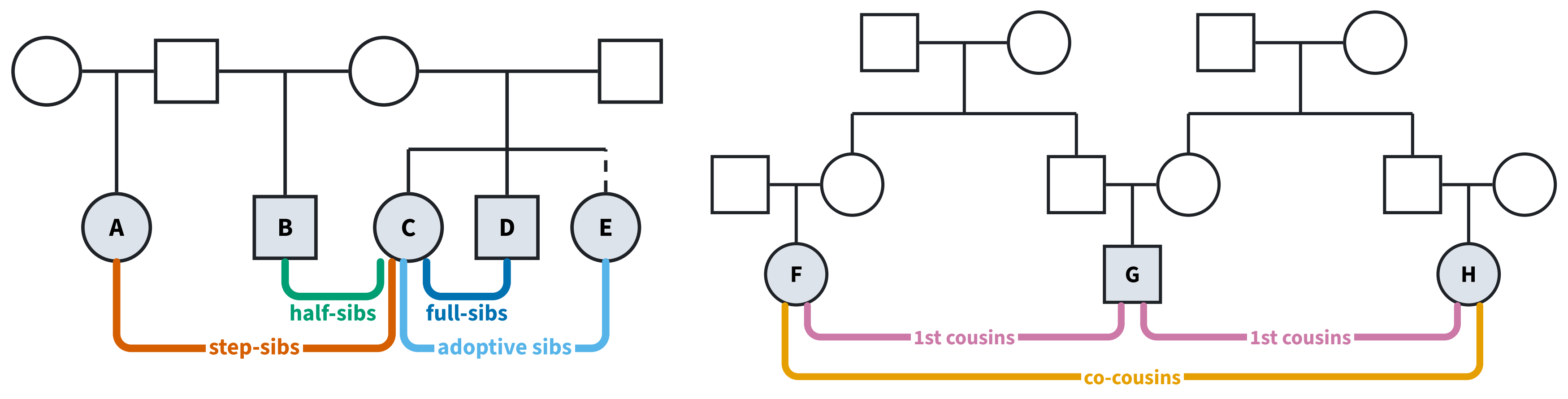
Supplementary Material

## Background

Relatives have similar test scores due to shared genetics and shared environments

[Eftedal et al. 2025, *PNAS*] fit the **Additive model** to school test scores of ∼1M Norwegian children

- identified the relative type of each pair of relatives
- fit the observed correlations by relative type as a function of the Additive model’s parameters



| relatives | $n$ | $f$ | correlation implied by Additive model |
|---|---|---|---|
| full siblings | 1 | 0 | $V_A\,(1+m)/2$ |
| half-siblings | 2 | 0.5 | $V_A\,(1+2m+rm)/4$ |
| step-siblings | 1+1 | 1 | $V_A\,m\,((1+r)/2)^2$ |
| adoptive siblings | – | 0 | $0$ |
| 1st cousins | 3 | 1 | $V_A\,((1+m)/2)^3$ |
| co-cousins | 2+2 | 2 | $V_A\,m\,((1+m)/2)^4$ |

### Two types of fits

| | Marginal Pair | Marginal Type |
|---|---|---|
| **unit** | each **pair** of relatives | each **relative type** |
| **weights** | $1/SE^2$ | uniform across types |
| **95% CI** | parametric bootstrap | leave-one-type-out jackknife |
| **use** | within-model comparisons | **across-model comparisons** |

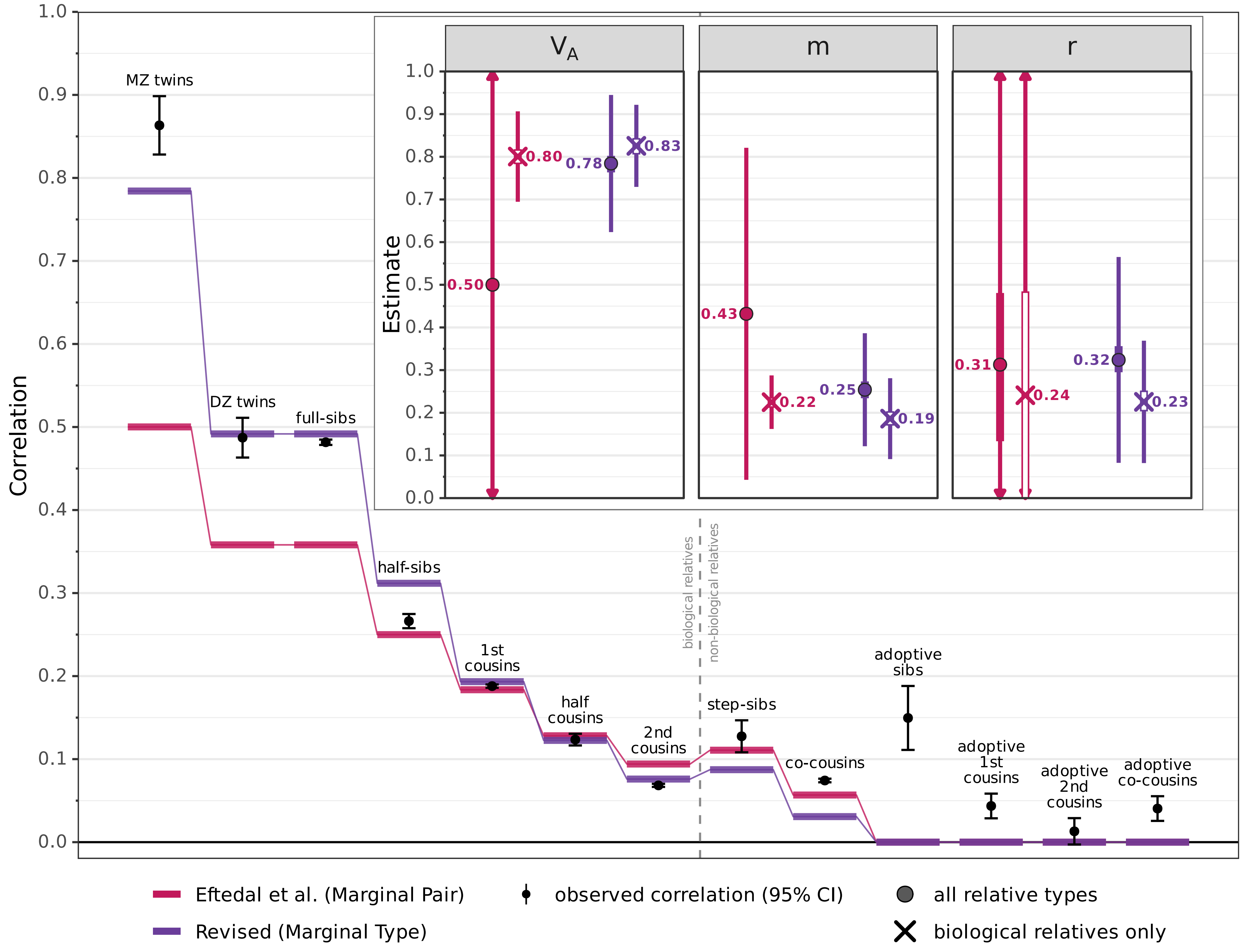
### Problems in Eftedal et al.’s estimator

- Only optimized for Marginal Pair, not Marginal Type
- Log-transformed correlations: over-weights distant relatives
- Incorrect inverse-variance weights: over-weight distant relatives
- Incorrect expected correlation for step-siblings
- Phenotypic ($r$) and genetic ($m$) correlations between mates estimated independently, although the model implies $m = V_A\,r$

## Revising Eftedal et al.’s results

### Additive model: Eftedal et al. vs. revised

The Revised estimator fixes errors in Eftedal et al.’s estimator and is optimized for the Marginal Type.



Colored bars denote each model’s predicted correlation for each relative type. Actual observed correlations shown in black. Eftedal et al.’s estimator optimizes for the Marginal Pair, while the Revised estimator optimizes for the Marginal Type.

Inset: parameter estimates compared between training cohorts, as done in Eftedal et al.: biological relatives only vs. all relative types. Both types of CIs are shown.

## Models with GxG and shared environment

### Gene–gene interactions (GxG)

Pairwise genetic interactions, not affected by assortative mating (AM).

$$\rho_k = \cdots + V_{\mathrm{GxG}}\left(\frac{1}{4}\right)^{n_k}$$

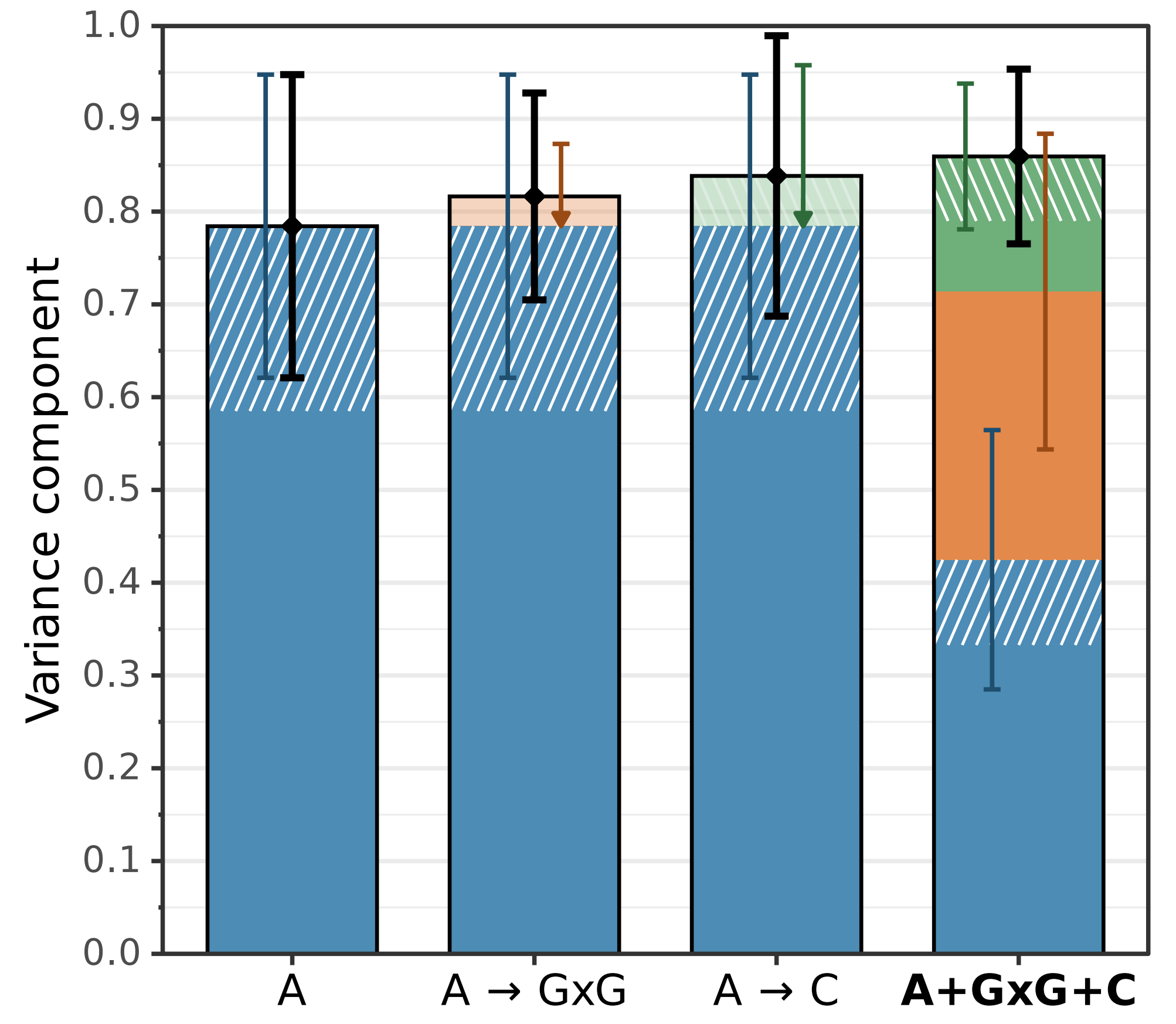
### Shared environment (C)

Environment shared across the extended family, decaying at a rate of $c$ with respect to family degree $f$:

$$\rho_k = \cdots + V_C\,c^{f_k}$$

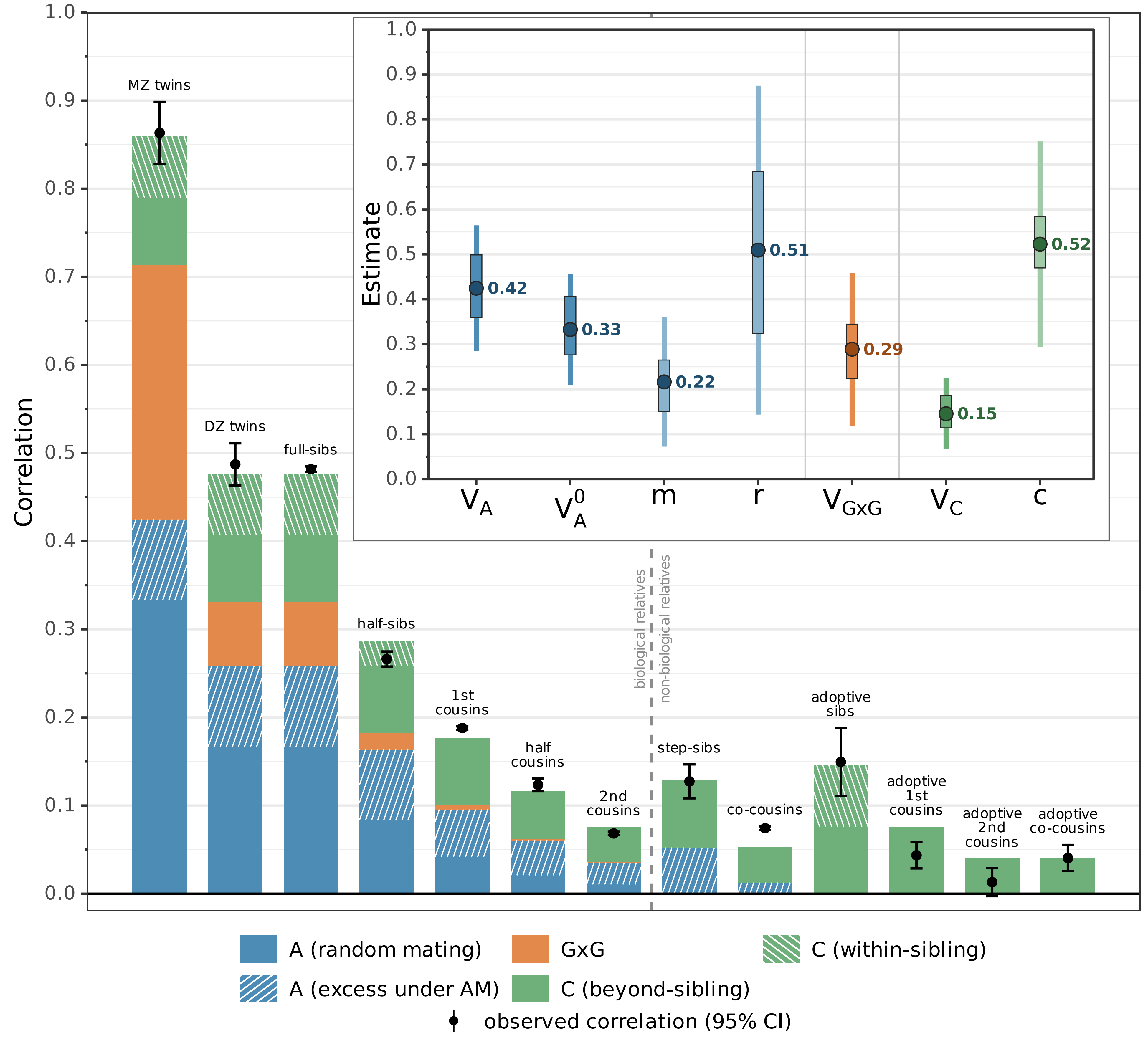
Can be decomposed into within-sibling ($V_C(1-c)$) vs. beyond-sibling ($V_C\,c$) shared environment.

### Variance components



A → B indicates two-step estimation, fixing A first. Marginal Type point estimates and 95% CIs. Color is faded if component variance is not significantly greater than 0. Black indicates sum of variance components. A+GxG+C is bolded and used as the example in the adjacent figure.

### Example: A+GxG+C



Predicted correlation of each relative type, decomposed by component. Marginal Type fit. Inset: parameter estimates; both types of CIs are shown.

## Key

| symbol | meaning |
|---|---|
| *Additive model* | |
| $n$ | genetic degree |
| $V_A$ | additive genetic variance |
| $V_A^0$ | additive variance under random mating, $V_A(1-m)$ |
| $m$ | mates’ genetic correlation |
| $r$ | mates’ phenotypic correlation |
| *Gene–gene interactions* | |
| $V_{\mathrm{GxG}}$ | GxG variance |
| *Shared environment* | |
| $V_C$ | shared-environment variance |
| $f$ | family degree |
| $c$ | decay per family degree |

## Takeaways

- Revising Eftedal et al.’s estimator for the Additive model yields **high** $V_A$ estimates that explain different relative types better
- Additive, GxG and shared-environment components are **collinear**: weak evidence for significant components beyond the Additive model
- No single model fully explains correlations across all relative types